

INHA UNIVERSITY TASHKENT

FALL SEMESTER 2024

SOC 3010

OPERATING SYSTEM

CREDITS/HOURS PER WEEK : 3/3

COURSE TYPE : TECHNICAL CORE SEQUENCE

INSTRUCTOR

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SOC 3010

OPERATING SYSTEM

LINUX KERNEL SOURCE INSTALLATION

Linux Kernal Source Installation

STEP 1 :

\$pwd

/home/user

\$sudo apt-get update

[sudo] password for user:

\$sudo apt-get install linux-source

[sudo] password for user:

Linux Kernal Source Installation

STEP 2 :

\$ls -la

```
drwxr-xr-x 27 root root 4096 Apr 21 10:28 linux-headers-4.4.0-21
drwxr-xr-x  7 root root 4096 Apr 21 10:28 linux-headers-4.4.0-21-generic
drwxr-xr-x  4 root root 4096 Apr 21 12:28 linux-source-4.4.0
lrwxrwxrwx  1 root root   45 Apr 21 13:12 linux-source-4.4.0.tar.bz2 -> linux-
source-4.4.0/linux-source-4.4.0.tar.bz2
```

USE THE SAME
VERSION HERE

\$tar xvjf /usr/src/linux-source-4.4.0.tar.bz2

(All Linux Kernel source files are untared(decompressed) and stored in your user directory under the **directory name - linux-source-X.X.0**)

\$ls CHECK YOUR VERSION OF LINUX SOURCE DOWNLOADED ON YOUR SYSTEM

Linux Kernal Source Installation

STEP 2 (contd):

\$cd linux-source-4.4.0

USE YOUR VERSION HERE

\$tree && tree -a > linux_directory_tree

\$ls

arch	dropped.txt	lib
block	firmware	MAINTAINERS
certs	fs	Makefile
COPYING	include	mm
CREDITS	init	
crypto	ipc	
debian	Kbuild	
Documentation	Kconfig	
drivers	kernel	

Using tree command you can generate the linux directory tree

Using pstree command you can generate the process tree

To install the tree command enter
\$sudo apt-get install tree

(Go to each directory - list all the source files in each directory and look at the content of c and h files using cat)

\$cat source_filename

Linux Kernel Source Code Structure

- ★ In order to find your way through the files of the source code, understanding the organization of the kernel directory tree is very important
- ★ As usual, all pathnames refer to the main directory of the Linux kernel, which is, in most Linux distributions, `/usr/src/linux`.
- ★ Linux source code for all supported architectures is contained in about 4500 C and Assembly files stored in about 270 subdirectories; it consists of about 2 million lines of code, which occupy more than 58 megabytes of disk space.

Linux Kernal Source Code Structure

□ Directory tree containing the Linux source code

init	- Kernel initialization code kernel
kernel	- core: processes, timing, program execution, signals, modules, . . .
mm	- Memory handling
arch	- Platform-dependent code
—i386	IBM's PC architecture
——kernel	Kernel core
——mm	Memory management

Linux Kernal Source Code Structure

❑ Directory tree containing the Linux source code

——math-emu	Software emulator for floating point unit
——lib	Hardware-dependent utility functions
——boot	Bootstrapping
——compressed	Compressed kernel handling
——tools	Programs to build compressed kernel image
—alpha	Compaq's Alpha architecture
—s390	IBM's System/390 architecture
—sparc	Sun's SPARC architecture
—sparc64	Sun's Ultra-SPARC architecture
—mips	Silicon Graphics' MIPS architecture
—ppc	Motorola-IBM's PowerPC-based architectures
—m68k	Motorola's MC680x0-based architecture
—arm	Architectures based on ARM processor

Linux Kernal Source Code Structure

□ Directory tree containing the Linux source code

fs

Filesystems

—proc	/proc virtual filesystem
—devpts	/dev/pts virtual filesystem
—ext2	Linux native Ext2 filesystem
—isofs	ISO9660 filesystem (CD-ROM)
—nfs	Network File System (NFS)
—nfsd	Integrated Network filesystem server
—fat	Common code for FAT-based filesystems
—msdos	Microsoft's MS-DOS filesystem
—vfat	Microsoft's Windows filesystem (VFAT)
—nls	Native Language Support

Linux Kernal Source Code Structure

□ Directory tree containing the Linux source code

—ntfs	Microsoft's Windows NT filesystem
—smbfs	Microsoft's Windows Server Message Block (SMB) filesystem
—umsdos	UMSDOS filesystem
—minix	MINIX filesystem
—hpfs	IBM's OS/2 filesystem
—sysv	System V, SCO, Xenix, Coherent, and Version 7 filesystem
—ncpfs	Novell's Netware Core Protocol (NCP)
—ufs	Unix BSD, SunOs, FreeBSD, NetBSD, OpenBSD, and NeXTStep filesystem
—affs	Amiga's Fast File System (FFS)
—coda	Coda network filesystem
—hfs	Apple's Macintosh filesystem
—adfs	Acorn Disc Filing System

Linux Kernal Source Code Structure

□ Directory tree containing the Linux source code

- efs SGI IRIX's EFS filesystem
- qnx4 Filesystem for QNX 4 OS
- romfs Small read-only filesystem
- autofs Directory automounter support
- lockd Remote file locking support

net **Networking code**

ipc **System V's Interprocess Communication**

Linux Kernal Source Code Structure

□ Directory tree containing the Linux source code

drivers

- block
- paride
- scsi
- char
- joystick
- ftape
- hfmodem
- ip2
- net
- sound
- video

Device drivers

- Block device drivers
- Support for accessing IDE devices from parallel port
- SCSI device drivers
- Character device drivers
- Joysticks
- Tape-streaming devices
- Ham radio devices
- IntelliPort's multiport serial controllers
- Network card devices
- Audio card devices
- Video card devices

❑ Directory tree containing the Linux source code

—cdrom	Proprietary CD-ROM devices (neither ATAPI nor SCSI)
—isdn	ISDN devices
—ap1000	Fujitsu's AP1000 devices
—macintosh	Apple's Macintosh devices
—sgi	Silicon Graphics' devices
—fc4	Fibre Channel devices
—acorn	Acorn's devices
—misc	Miscellaneous devices
—pnp	Plug-and-play support
—usb	Universal Serial Bus (USB) support
—pci	PCI bus support
—sbus	Sun's SPARC SBus support
—nubus	Apple's Macintosh Nubus support
—zorro	Amiga's Zorro bus support
—dio	Hewlett-Packard's HP300 DIO bus support
—tc	Sun's Turbo Channel support

Linux Kernal Source Code Structure

□ Directory tree containing the Linux source code

lib	General-purpose kernel functions
include	Header files (.h)
—linux	Kernel core
——lockd	Remote file locking
——nfsd	Integrated Network File Server
——sunrpc	Sun's Remote Procedure Call
——byteorder	Byte-swapping functions
——modules	Module support
—asm-generic	Platform-independent low-level header files
—asm-i386	IBM's PC architecture
—asm-alpha	Compaq's Alpha architecture
—asm-mips	Silicon Graphics' MIPS architecture
—asm-m68k	Motorola-IBM's PowerPC-based architectures

Linux Kernal Source Code Structure

❑ Directory tree containing the Linux source code

—asm-ppc	Motorola-IBM's PowerPC architecture
—asm-s390	IBM's System/390 architecture
—asm-sparc	Sun's SPARC architecture
—asm-sparc64	Sun's Ultra-SPARC architecture
—asm-arm	Architectures based on ARM processor
—net	Networking
—scsi	SCSI support
—video	Video card support
—config	Header files containing the macros that define the kernel configuration

scripts External programs for building the kernel image

Documentation Text files with general explanations and hints about kernel components

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WISH YOU ALL THE BEST

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